

NON-ODONTOGENIC EPITHELIAL AND CONNECTIVE TISSUE TUMORS OF ORAL CAVITY

Dr. Sajda Khan Gajdhar
MDS, Oral and Maxillofacial Pathology



LEARNING OUTCOMES

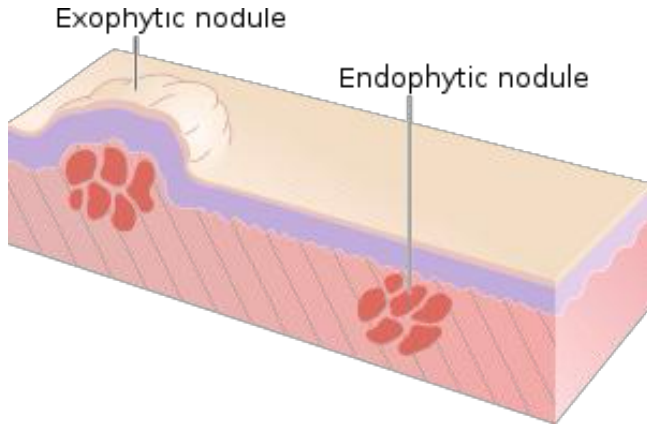
By the end of this chapter students will be able to:

1. Classify the **benign and malignant tumors** of oral cavity.
2. Describe the **origin of common benign and malignant** tumors of oral cavity.
3. Explain the **etiopathogenesis, clinicopathologic features and management** of **squamous papilloma** and **squamous cell carcinoma**.
4. Correlate the **clinical and microscopic features of common benign and malignant tumors** of oral cavity.

Neoplasia: (Willis Definition) is defined as abnormal, uncoordinated, purposeless proliferation of tissue, the growth of which doesn't regress after removal of the stimuli which evoked the change.



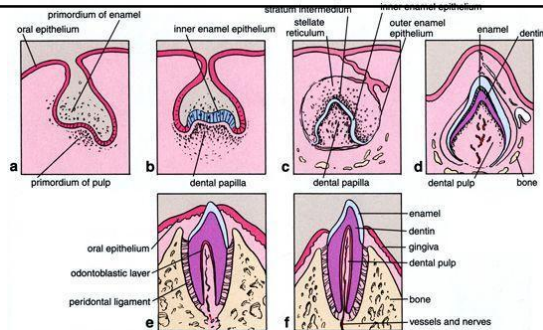
BENIGN VS MALIGNANT



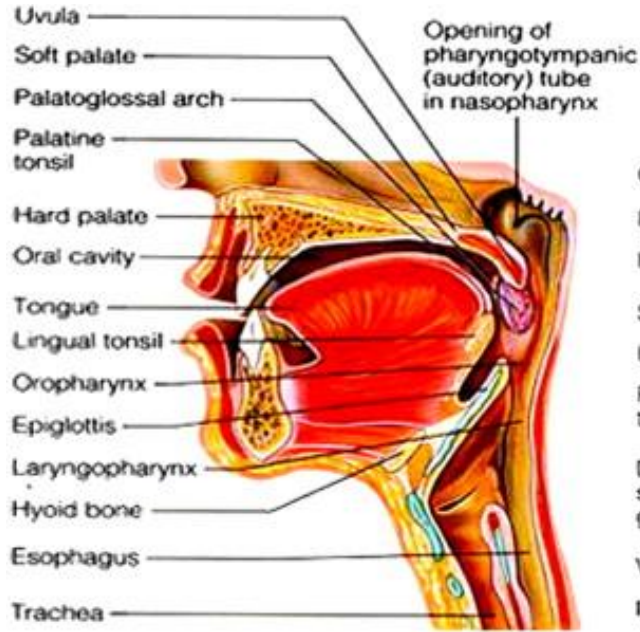
Benign tumors	Malignant tumor
Slowly growing mass	Rapidly growing mass
Regular surface, capsulated, not attached to deep structures	Irregular surfaces, Non-capsulated attached to deep structures
Noninvasive to another organ or tissues	Invasive to other organs
No spread or metastasis	Spread and metastasis
Well differentiated all the them	Poorly differentiated, moderately or well differentiated
No recurrence after surgery	Recurrence after surgery
No bleeding in cut surfaces	Bleeding from cut surfaces is common
Named by adding suffix -oma	Named by adding suffix sarcoma or carcinoma
Slight pressure effect on the neighboring organ	Remarkable pressure effect on neighboring tissue

ODONTOGENIC VS NON-ODONTOGENIC

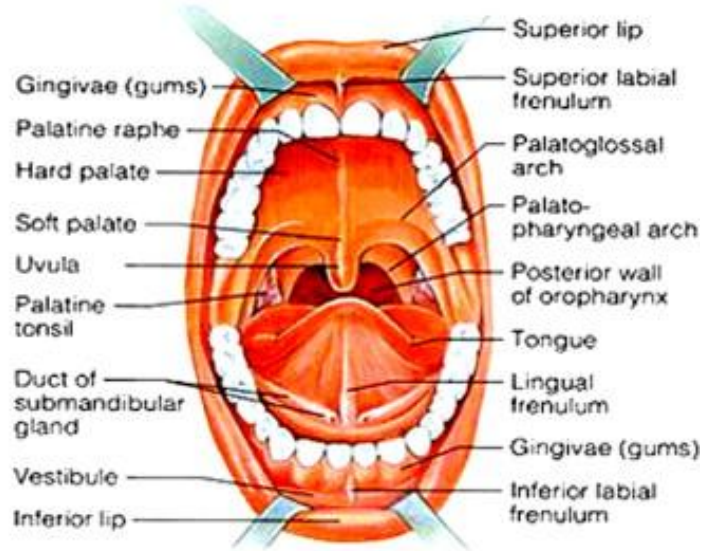
- Oral tumors are broadly divided into **odontogenic** (from tooth-forming tissues) and **non-odontogenic**.
- Non-odontogenic tumors arise from **epithelium, connective tissue, muscle, nerve, or vascular tissue**.
- Understanding them helps differentiate **benign vs malignant lesions**, plan treatment, and anticipate prognosis.



ODONTOGENIC VS NON-ODONTOGENIC



(a)



(b)

CLASSIFICATION

EPITHELIUM DERIVED

- **Papilloma – epithelium derived**
- **Focal Epithelial Hyperplasia - epithelium derived**
- **Keratoacanthoma - epithelium derived**
- ***Nevus*- epithelium derived**

CONNECTIVE TISSUE DERIVED

-
- **Fibrous hyperplasia - connective tissue derived**
 - **Angiomas- blood, lymph vessel derived**
 - **Neuroma - neural derived**
 - **Neurofibroma - neural derived**
 - **Neurilemmoma- neural derived**
 - **Lipoma- Adipose tissue derived**
 - **Leomyoma & Rhabdomyoma - muscle derived**

**PREMALIGNANT
LESIONS/CONDITIONS**

- Leukoplakia- epithelium derived
- *Lichen planus- epithelium derived*
- Carcinoma in situ- epithelium derived
- Oral submucous fibrosis - ept & c.t. derived
- Erythroplakia- epithelium derived
- Plummer Vinson syndrome- Fe deficiency

MALIGNANT LESIONS

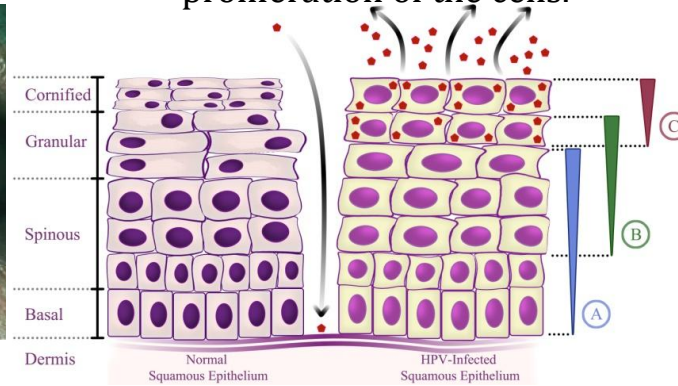
- **Oral Squamous cell carcinoma- ept. derived**
- Fibrosarcoma- connective tissue derived
- *Verrucous carcinoma- ept derived*
- *Malignant melanoma- epit derived*
- Liposarcoma- Adipose tissue derived
- Leomyosarcoma & Rhabdomyosarcoma - muscle derived
- Malignant Peripheral Nerve Sheath Tumor- nerve derived

Non-Odontogenic Epithelial Tumors

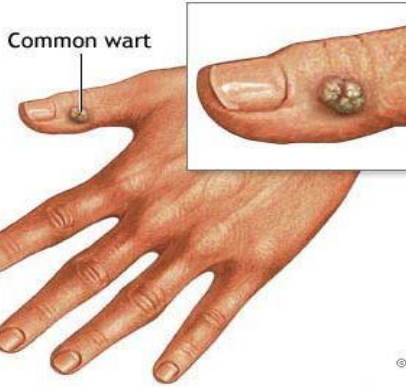
SQUAMOUS PAPILLOMA (common wart)



- A **benign epithelial tumor**.
- HPV - DNA virus. **Exofighting**
- The virus replicates inside the nucleus of the human keratinocytes.
- Affects basal keratinocytes and leads to proliferation of the cells.



Clinical Features



- Very common lesion approximating **2.5% of oral soft tissue** lesions.
- Usually lesser **than 1cm** in greatest dimension
- Common site skin (hands, face and neck)
- Oral cavity-**tongue, buccal mucosa**
- **White exophytic granular or cauliflower like growth**
- Solitary are common but multiple lesions are also seen.
- **HPV 16 & 18 associated** with neoplastic changes common in HIV patients.
- Diagnosis is confirmed by biopsy.



Finger like projections/papillae

PAPILLOMATOSIS

Fibrovascular core

Increase in keratin layer

HYPERKERATOSIS

Increase in the number of layers

ACANTHOSIS

Enlarged squamous cell with
clear cytoplasm with raisin like
nuclei

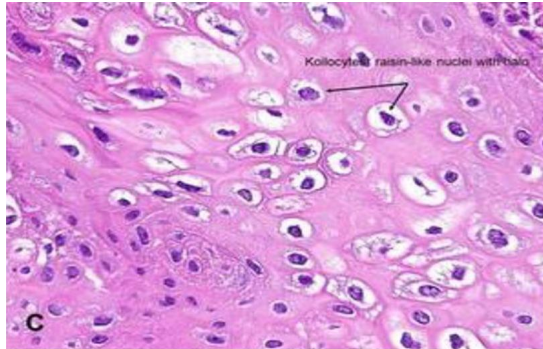
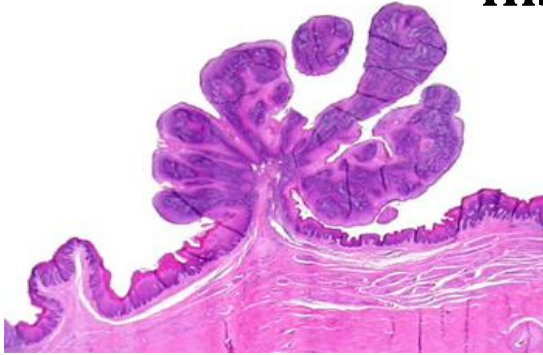


KOILOCYTE

Nuclear membrane - Wrinkled
appearance - **RAISIN**
(partially dried grape)

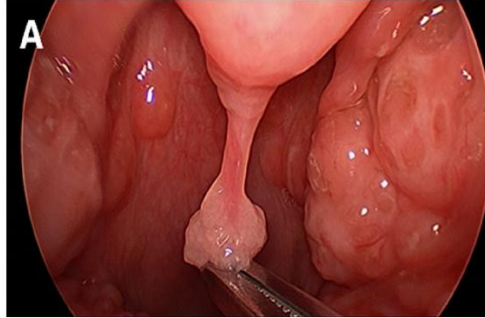


Histopathology



- Benign epithelium with minor amounts of connective tissue elements.
- Finger like extensions of epithelium
- Vascularized connective tissue core underlying epithelium.
- Koilocytes – Pyknotic nuclei surrounded by edematous or optically clear zone.
- Few lesions may show dysplastic features??

Management

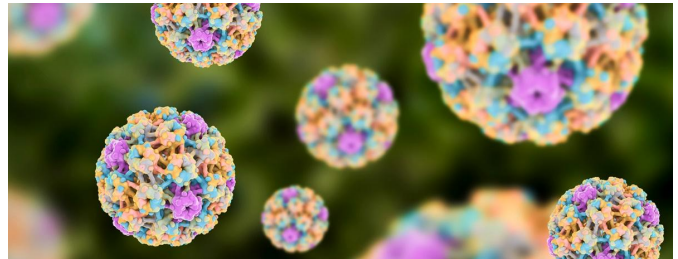


- Often no treatment
- Surgical excision
- Laser ablation
- Uncommon reoccurrence unless the immunity is low.

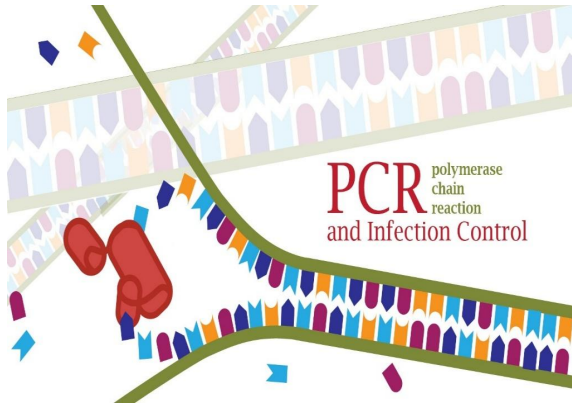
FOCAL EPITHELIAL HYPERPLASIA



- Also known as **Heck's disease**, **Multifocal papillomavirus epithelial hyperplasia**.
- Caused by human papillomavirus (**HPV – 13 & 32**)
- Frequent in **younger age groups** and **sometimes has also a characteristic familial occurrence and common in HIV Patients** .

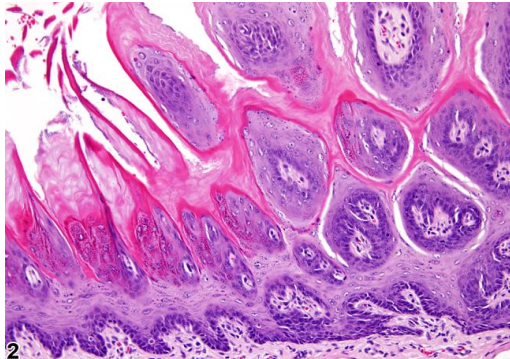
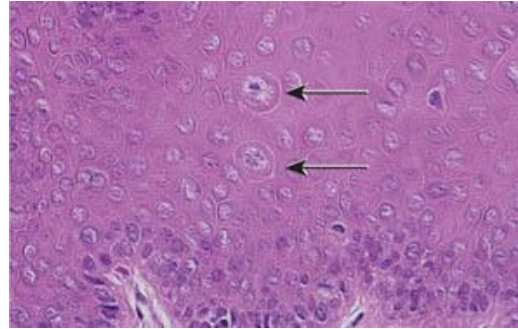


Clinical Features and Diagnosis



- **Multiple soft papules of whitish or normal color** with a smooth surface or papillary and measuring 1 to 10 mm in diameter.
- The lesions are **well demarcated, painless, tend to disappear spontaneously, and are predominantly found on the lower lip, buccal mucosa and tongue.**
- **Polymerase chain reaction (PCR)** and sequencing of PCR products is used for diagnosis.

Histopathology



- Acanthosis and Parakeratosis
- **Clubbing and fusion of epithelial ridges**
- **Mitosoid bodies seen in superficial keratinocytes** (Cells resembling mitotic appearance)-due to collapsed nucleus.

Management



- Spontaneous regression due to immune response.
- Multiple lesions
 - Observation
- Single lesions
 - Surgical excision/Laser ablation

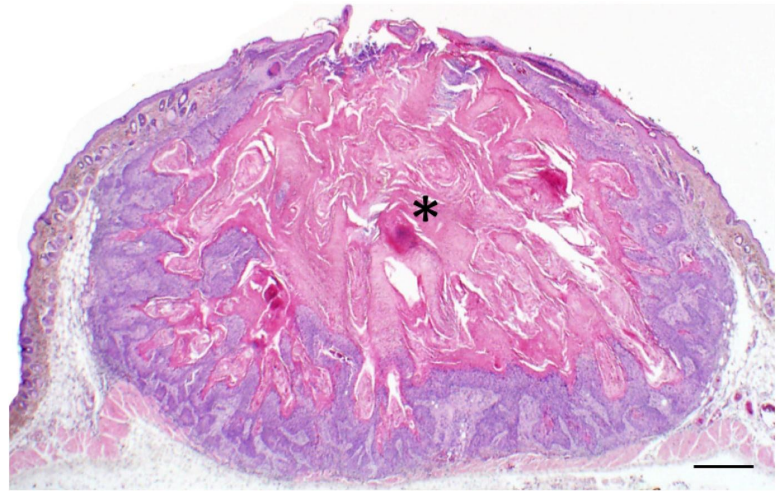
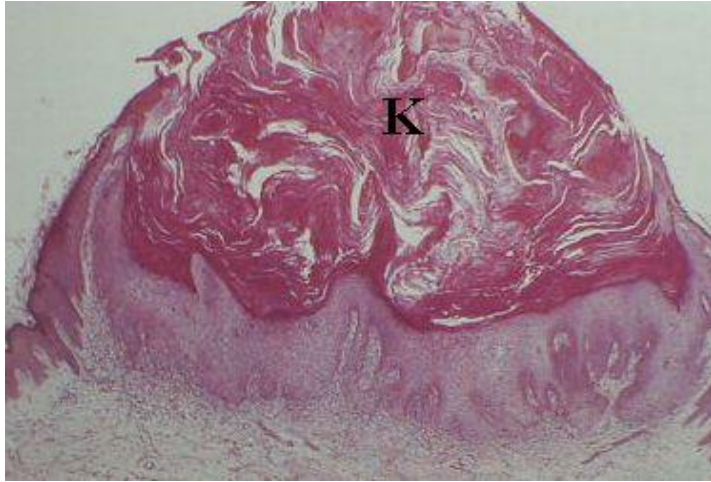
Keratoacanthoma



- Benign epithelial neoplasm
- common low-grade (unlikely to metastasize or invade) skin tumor
- originates in the hair follicles epithelium above the sebaceous glands
- Rapidly growing skin cancer



Management: Surgical excision; may regress spontaneously.



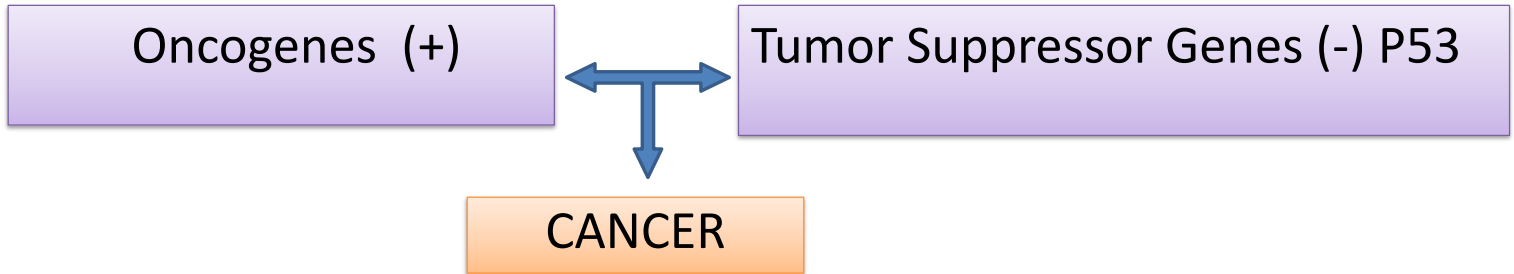
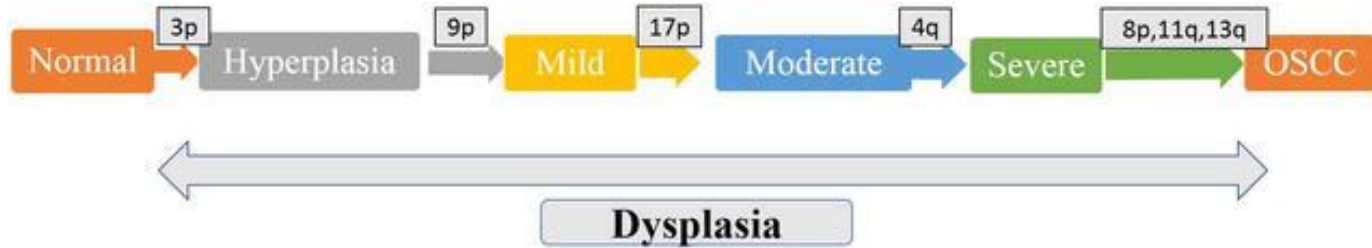
well-circumscribed, symmetric crater **filled with keratin**, with proliferation of squamous epithelium at the edges.

ORAL SQUAMOUS CELL CARCINOMA



- Oral squamous cell carcinoma (OSCC) is **commonest malignancy affecting oral cavity**
- About 95% of oral cancers are OSCCs.
- Other 5% of cancers of oral cavity are melanoma, lymphoma, metastatic cancer of other parts of body, basal cell carcinoma, fibrosarcoma, osteosarcoma, & odontogenic malignancies.

ETIOPATHOGENESIS



Progression of OSCC

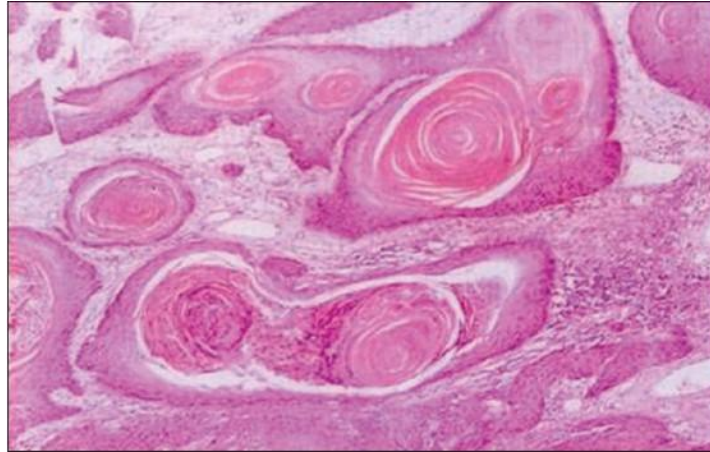
- **Loss of cell cycle control:**

Increased proliferation and reduced apoptosis

- **Increased neoplastic cell motility:**

Invasion and metastasis

Invasion and Metastasis



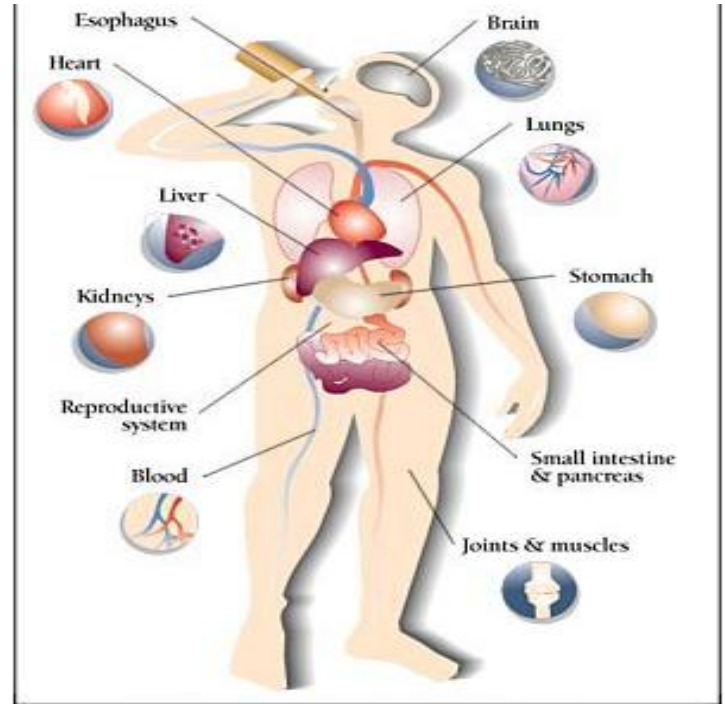
ETIOLOGY: Smoking Tobacco



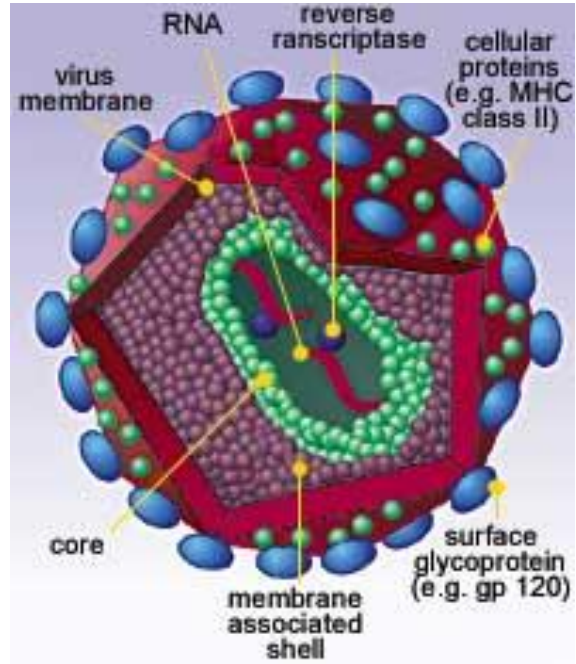
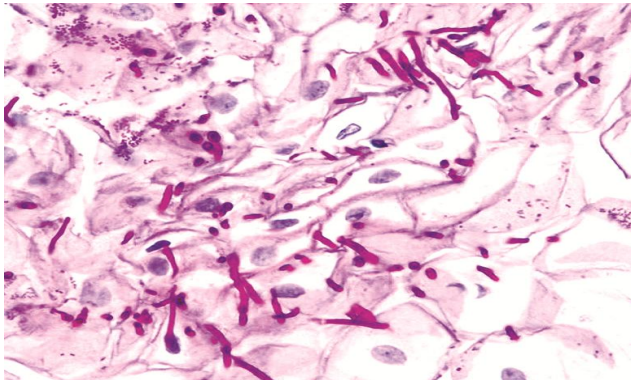
Smokeless Tobacco



Spirit - Alcohol



Sepsis



Syphilis



Spices



Sunlight



Sharp Tooth – Chronic Trauma

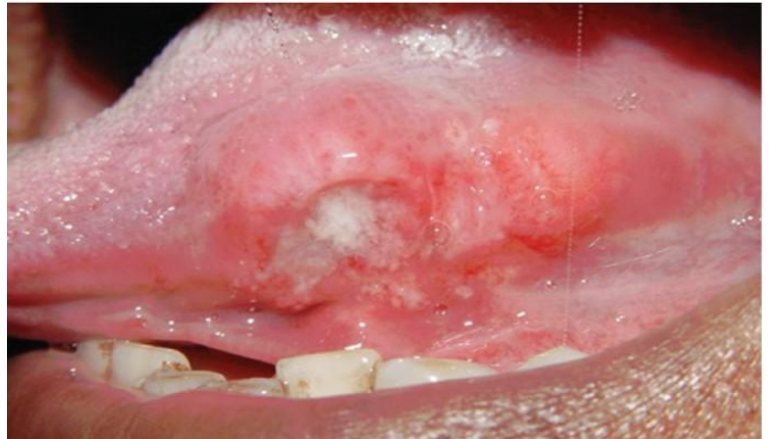


CLINICAL FEATURES

Ca - Lip



Ca - Tongue



Ca - Floor of mouth



Ca - Buccal mucosa



Grading and staging

- TNM Classification
- STNMP Classification
 - Borders Grading

TNM Staging System ..

- **T**  **Tumor**
- **N**  **Nodal status**
- **M**  **Metastasis**

Classification:

- **Primary tumor, as follows:**

T1 - Tumor 2 cm or smaller

T2 - Tumor 2-4 cm

T3 - Tumor larger than 4 cm

T4 - Tumor larger than 4 cm and deep invasion to muscle, bone, or deep structures

- Lymphatic node involvement, as follows:

N0 - No nodes

N1 - Single homolateral node smaller than 3 cm

N2 - Nodes homolateral 3-6 cm or bilateral

N3 - Nodes larger than 6 cm

- Tumor metastasis, as follows:

M0 - No metastasis

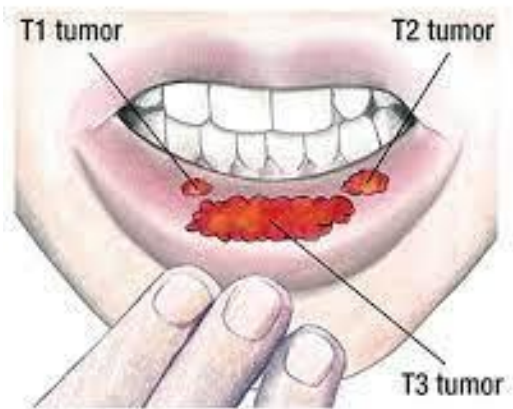
M1 - Metastasis noted

Stage I



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Stage III



Staging

STAGE			
Stage I	T ₁ , No, Mo		
Stage II	T ₂ , No, Mo.		
Stage III	T ₃ , No, Mo	T ₁ , T ₂ , T ₃ , N ₁ , Mo	
Stage IV	T ₄ , No, Mo	Any T, N ₂ or N ₃ , Mo	Any T, any N, any M

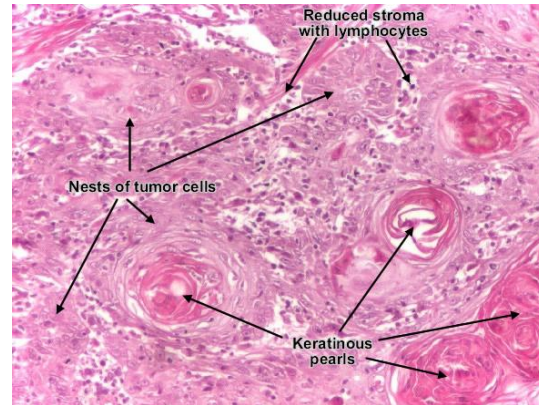
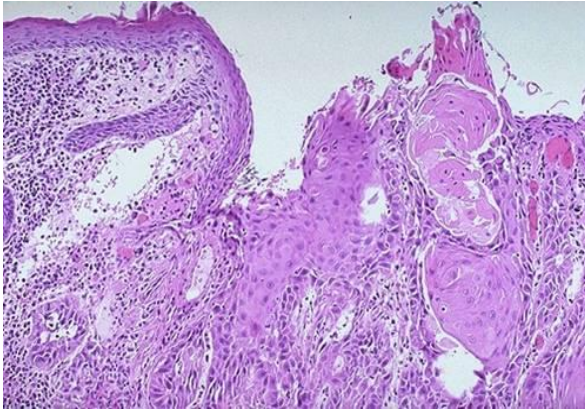
TNM CLASSIFICATION???????

- T₃, No, Mo
- T₄, No, Mo
- T₁, No, Mo
- T₂, No, Mo
- T₁, No, M₁
- T₁, N₁, M₁



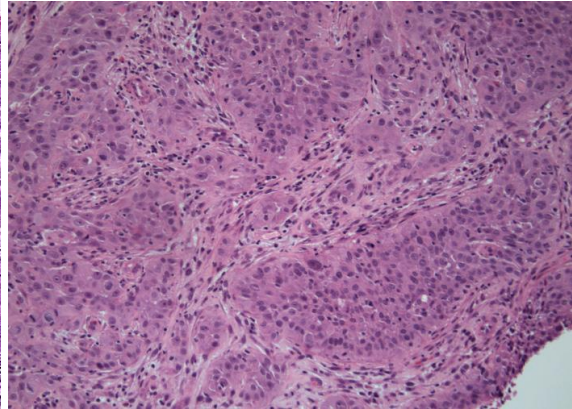
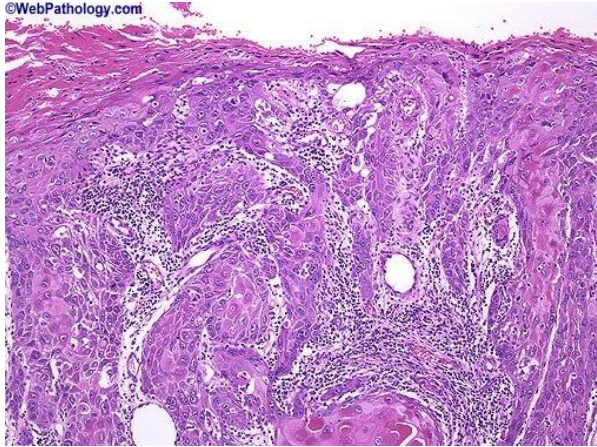
BRODER'S HISTOPATHOLOGY STAGING SYSTEM

Well differentiated SCC

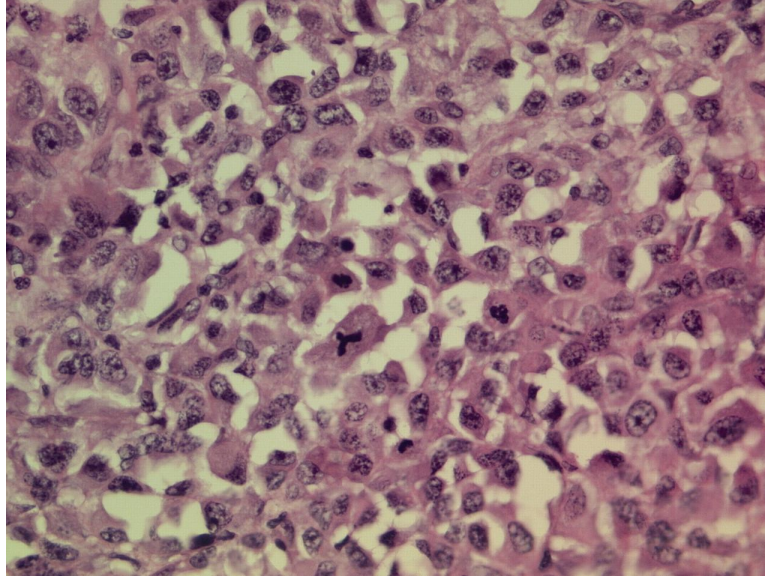


MODERATELY DIFFERENTIATED SCC

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POORLY DIFFERENTIATED SCC



HISTOPATHOLOGIC VARIANTS

- Verrucous carcinoma
- Spindle cell carcinoma/Sarcomatoid Carcinoma
- Basaloid squamous cell carcinoma.

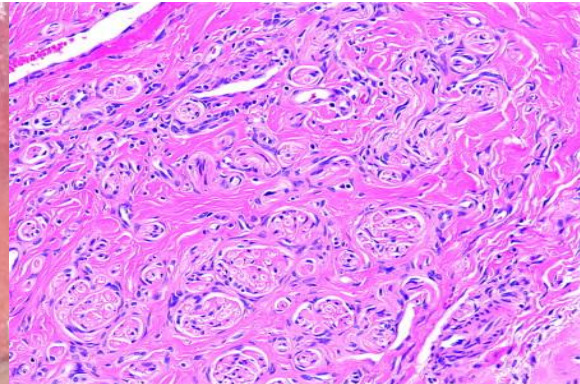
TREATMENT

TNM CLASSIFICATION + PATHOLOGIC GRADING

- Surgery
- Radiation
- Chemotherapy

Non-Odontogenic Connective Tissue Tumors

Neuroma

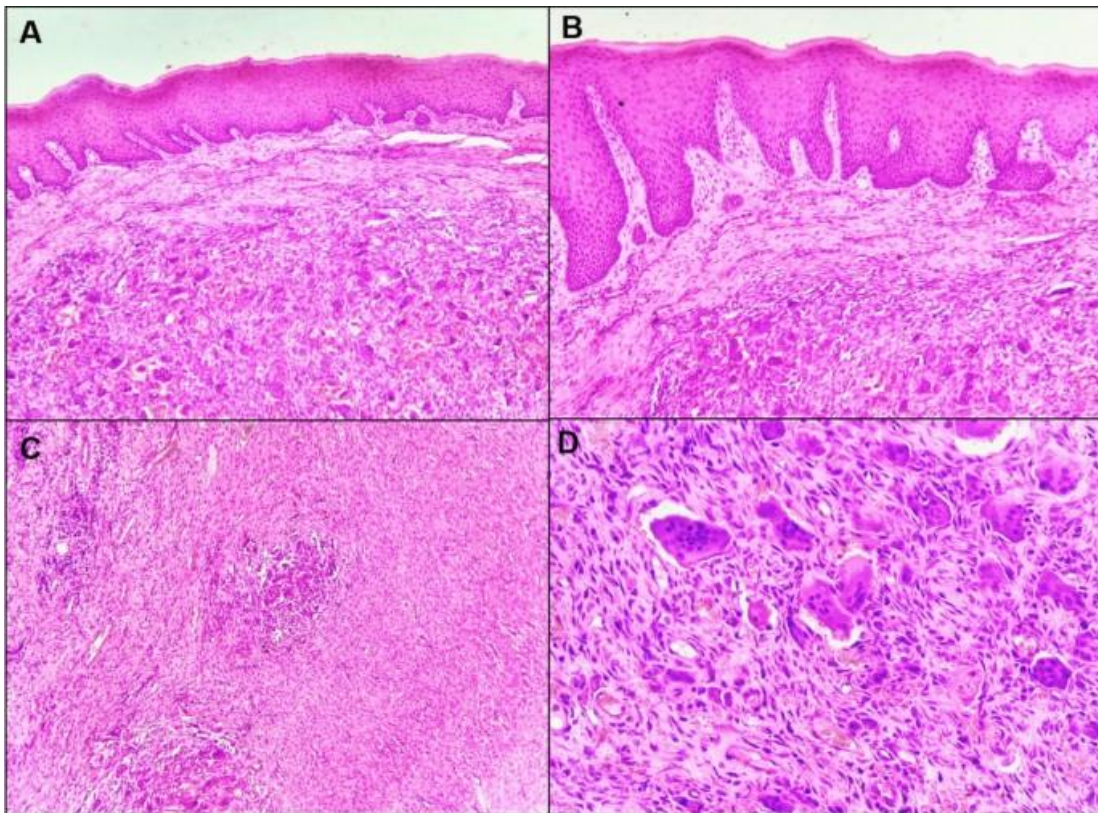


- Uncommon, benign tumors.
- Types: Traumatic neuroma, Solitary circumscribed neuroma, Genetic syndrome associated neuroma.
- **Well circumscribed neural proliferation** that may demonstrate haphazard growth.
- **Site: Buccal mucosa, tongue, lips, palate and gingiva.**
- Solitary circumscribed neuromas result from a **proliferation of Schwann cells within the nerve.**

Peripheral Giant Cell Granuloma (PGCG)

- **Definition:** Reactive lesion from periodontal ligament or periosteum.
- **Clinical features:** Red-blue nodular lesion on **gingiva or alveolar ridge**.
- Often triggered by **local trauma or irritation**, it primarily affects adults in their 5th-6th decade, slightly more often in women,
- **Histopathology:**
 - Multinucleated giant cells in vascular stroma
 - Hemosiderin deposits
- **Management:** Surgical excision with curettage.





REFERENCES

1. Regezi and Sciubba, clinicopathologic correlations
2. Shafer's textbook of oral pathology
3. Neville text book of oral and maxillofacial pathology